

Maximum Marks:20

4. If the sum of n observations is 630 and their Arithmetic mean is 42, then the value of n is _____.
- (i) 21 (iii) 15
(ii) 30 (iv) 20
5. Which of the measure of central tendency is not affected by extreme values? _____.
- (i) arithmetic mean (iii) harmonic mean
(ii) median (iv) geometric mean
6. The Arithmetic mean of seven observations is 8. A new observation 16 is added. The mean of eight observations is _____.
- (i) 12 (iii) 9
(ii) 8 (iv) 24
7. The data obtained in an experiment on total body calcium of a patient with osteoporosis (nearest grams) is of type _____.
- (i) Ordinal (iii) Discrete
(ii) Nominal (iv) Continuous
8. The Geometric mean of two numbers $\frac{1}{16}$ and $\frac{4}{25}$ is _____.
- (i) $\frac{1}{10}$ (iii) $\frac{1}{100}$
(ii) 10 (iv) 100
9. A frequency distribution of Heart Rate of 30 subjects have class limits 70-82,83-95,....The mid point of first class interval is equal to _____.
- (i) 82.5. (iii) 76.
(ii) 0.5. (iv) 12.
10. The standard deviation of variable is _____.
- (i) Mean squared deviation of variable about mean (iii) Mean deviation about mean
(ii) Root mean squared deviation about Median (iv) Root mean squared deviation about mean
11. For n observations $x_1, x_2, \dots, x_n$, the sum of squares of deviations is minimum when deviations are taken from _____.
- (i) mean (iii) mode
(ii) median (iv) geometric mean

12. Let S_1 and S_2 are standard deviations of series 2,5,8,11 and 1,4,7,10, respectively then _____.

(i) $S_1 > S_2$

(iii) $S_1 < S_2$

(ii) $S_1 = S_2$

(iv) $S_1 = S_2 + 1$

13. Which of the following is a linear function(model)? _____.

(i) $Y = \beta_0 + e^{\beta_1 X} + \varepsilon$

(iii) $Y = e^{\beta_0 + \beta_1 X + \varepsilon}$

(ii) $Y = \beta_0 + \beta_1 X + \varepsilon$

(iv) Both (i) and (iii)

14. The slope of the regression line $Y = 2 \left(\frac{1-x}{3} \right)$ is _____.

(i) -2

(iii) $-\frac{2}{3}$

(ii) $-\frac{1}{3}$

(iv) $\frac{2}{3}$

15. The simple correlation between variables X and Y that is $r_{xy} = 0.8$, if $U = 2X, V = -3Y$ then $r_{uv} =$ _____.

(i) -0.8

(iii) 0.8

(ii) -0.48

(iv) 0.48

16. The variable Y in a regression model $Y = \beta_0 + \beta_1 X + \varepsilon$ is known as _____.

(i) independent variable

(iii) predictor variable

(ii) explanatory variable

(iv) response variable

17. The simple correlation coefficient between variables X and Y is known to be zero. We then can conclude that _____.

(i) the covariance between X and Y is zero

(iii) Both (i) and (ii)

(ii) there exists no linear relationship between X and Y

(iv) the variances of X and Y are equal

18. The least squares estimate $\hat{\beta}_1$ based on n pairs $(x_i, y_i), i = 1, 2, \dots, n$ of observations in regression model $Y = \beta_0 + \beta_1 X + \varepsilon$ is equal to _____.

(i) $\frac{\sum_{i=1}^n (y_i - \bar{y})(x_i - \bar{x})}{\sum_{i=1}^n (x_i - \bar{x})^2}$

(iii) $\frac{Cov(x, y)}{V(x)}$

(ii) $\frac{\bar{y} - \hat{\beta}_0}{\bar{x}}$

(iv) Both (i) and (iii)

19. A student obtained simple regression equation $\hat{Y} = -3.6 + 3.1X$ where X is diameter(inch) of tree trunk and Y is tree height. The estimate of the average height of all trees having a trunk diameter of 7 inches is _____.

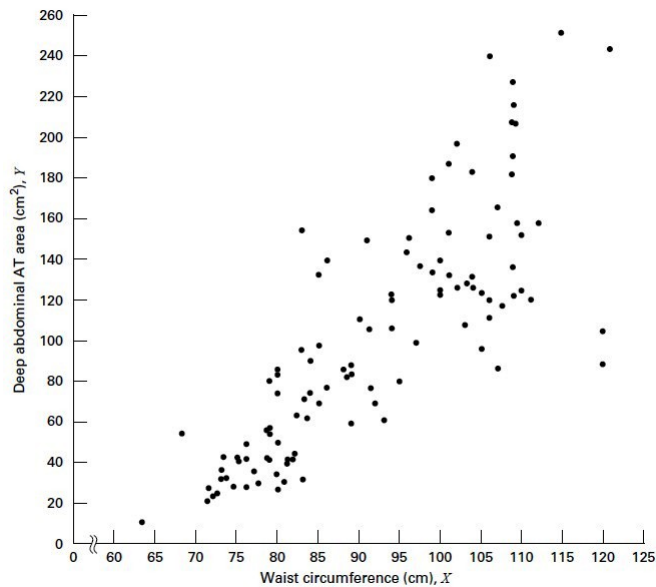
(i) 25.3

(iii) 21.7

(ii) 18.1

(iv) 3.42

20. A researcher obtained the scatter diagram of the measurements on deep abdominal AT (Y) obtained by CT and waist circumference (X) of 109 men between the ages of 18 and 42 years who were free from metabolic disease that would require treatment.



From the scatter plot one may conclude that the relationship between X and Y is _____.

(i) linear and positive

(iii) nonlinear and positive

(ii) linear and negative

(iv) nonlinear and negative

End of MCQ

CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY
Second Semester of BSc Examination May 2018
BS211 Biostatistics

Date: 04/05/2018, Friday

Time: 10:30 a.m. to 01:00 p.m.

Maximum Marks: 50

Instructions:

- Section I and II must be attempted in TWO ANSWER SHEET.
- Make suitable assumptions and draw neat figures wherever required.
- Use of non-programmable calculator is allowed.
- Show necessary calculations.
- Graph paper will be provided on demand.

SECTION I

Q - II Answer the following questions as directed. [20]

1. Define the following terms. [08]
 - (a) Biostatistics.
 - (b) Population in Biostatistics with illustration.
 - (c) Arithmetic mean and one mathematical property.
 - (d) Geometric mean and harmonic mean.
2. Attempt any three questions. [12]
 - (i) Data below shows the lengths (in cm) of 10 leaves from an ornamental bush measured by a researcher. Compute

Table 1: lengths (in cm)

1.6	1.9	2.2	2.1	2.2	1.0	0.8	0.6	1.1	2.2
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- (i) Sample mean (ii) Quartiles Q_1 , Q_2 and Q_3
- (ii) The Table 2 shows the age distribution of live births in Albany County, New York, for year 2000. Compute the Mode Age.

Table 2: Live Births by Resident County and Mother Age.

Mother's Age	15 - 19	20 - 24	25 - 29	30 - 34	35 - 39	40 - 44	45 - 49
Number of Live Births	258	585	841	981	526	99	4

- (iii) The Table 3 shows frequency distribution of systolic blood pressure of 199 subjects. Construct the Histogram and Ogive from the frequency distribution.

Table 3: Frequency distribution: systolic blood pressures (mmHg)

Systolic BP	85-94	95-104	105-114	115-124	125-134	135 -144	145-154	155-164	165-174
Number of Patients	6	20	27	48	34	36	17	5	6

- (iv) Compute the Median systolic blood pressure from the frequency data in Table 3.

SECTION II

Q - III Answer the following questions. **[30]**

1. Explain in brief **[12]**

- (i) Standard deviation and one mathematical property. (iii) Types of Variables in Regression Model.
(ii) Mean deviation. (iv) Scatter diagram.

2. Attempt any three questions. **[18]**

- (i) The data come from a study that investigated optimal conditions for the extraction of tobacco alkaloids using ultrasonic and microwave extraction methods. The first row shows the extraction temperature ($^{\circ}\text{C}$), while the second row gives the percentage nicotine extracted from tobacco.

Table 4: The nicotine study data

Temperature $^{\circ}\text{C}$	41	41	74	74	30	85	57.5	57.5
% Nicotine	3.279	3.401	3.973	4.319	3.145	4.595	3.945	4.243

- (a) Identify dependent and independent variables from the data (Table 4) (b) Assume a linear relationship, Predict the % Nicotine at Temperature 50°C from regression line.
(ii) In medicine manufacturing process, mainly two processes viz. process -X and process -Y, are performed in a company which has cause and effect relationship. Following data is of time (in minutes) required to complete process X and process Y.

Table 5: time in minutes

X	65	66	67	67	68	69	70	72
Y	67	68	65	72	72	72	69	71

- (a) Construct the scatter plot of variables in table (5)
(b) Find simple linear correlation r between the variables in table (5) and write your conclusion about value of r .
(iii) A study was conducted to demonstrate that soy beans inoculated with nitrogen-fixing bacteria yield more and grow adequately without expensive environmentally deleterious synthesized fertilizers. There were 8 inoculated plants (I) and 8 uninoculated plants (U). The plant yield as measured by pod weight for each plant is given in Table 6.

Table 6: Pod weight (g) from inoculated (I) and uninoculated (U) plants

Sample	1	2	3	4	5	6	7	8
I	1.76	1.45	1.03	1.53	2.34	1.96	1.79	1.21
U	0.49	0.85	1.00	1.54	1.01	0.75	2.11	0.92

Compute the standard deviations of pod weight for both inoculated (I) and uninoculated (U) plants and write your comment based on your computations.

- (iv) Table 7 displays the frequency distribution of IQ scores of 60 fifth-grade students chosen at random from one school.

Table 7: Frequency distribution: IQ test scores for 60 randomly chosen fifth-grade students

Lower Limit	75	85	95	105	115	125	135	145
Upper Limit	84	94	104	114	124	134	144	154
Number of Students	2	3	10	16	13	10	5	1

Compute the standard deviation of IQ score from the frequency distribution.
(Table 7)

- (v) Table 8 displays the IQ scores of 60 fifth-grade students chosen at random from one school.

Table 8: IQ test scores for 60 randomly chosen fifth-grade students

StudentNo	1	2	3	4	5	6	7	8	9	10
IQScore	145	139	126	122	125	130	96	110	118	118
StudentNo	11	12	13	14	15	16	17	18	19	20
IQScore	101	142	134	124	112	109	134	113	81	113
StudentNo	21	22	23	24	25	26	27	28	29	30
IQScore	123	94	100	136	109	131	117	110	127	124
StudentNo	31	32	33	34	35	36	37	38	39	40
IQScore	106	124	115	133	116	102	127	117	109	137
StudentNo	41	42	43	44	45	46	47	48	49	50
IQScore	117	90	103	114	139	101	122	105	97	89
StudentNo	51	52	53	54	55	56	57	58	59	60
IQScore	102	108	110	128	114	112	114	102	82	101

- (a) Select again a random sample of 9 students (without replacement) from the data (Table 8) using remainder approach.
(b) Obtain the IQ Scores of students selected by you in question (a).
(c) Compute the Mean deviation about **Median** of IQ scores of sample obtained by you.
Use Table of Random Numbers given below.

Table 9: Random Numbers: **Read from left to right**

11	16	43	63	18	75	6	13	76
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